ACADEMY OF TECHNOLOGY
Internal Examination (CA2), Odd Semester, 2026

Programme/Discipline: B.Tech/ CSE **Semester:5th**
Paper Name: Operating Systems **Paper Code:** PCC-CS502 **UPID:** 5507
Date of Examination: 06/10/2026 **Duration:** 1 Hour **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1. vii	2	3	4	5	6	7
CO Mapping	CO3	CO4	CO4	CO3	CO3	CO3	CO4	CO3	CO3	CO4	CO4	CO4	CO3
Bloom's Level	L1	L1	L1	L1	L2	L1	L1	L2	L2	L2	L3	L4	L4

Group A

[1 × 5 = 5]

1. **Attempt any five questions from the following.**

i. **A binary semaphore can take the values**
a) 0 and 1 only b) 0 to N *(N denotes the number of processes)* c) any integer d) −1 and +1

ii. **In a resource-allocation graph, an edge drawn from a resource box to a process circle is**
a) a claim edge b) a request edge c) an assignment edge d) a wait-for edge

iii. **Which of the following is not one of the four necessary conditions for deadlock?**
a) Mutual exclusion b) Hold and Wait c) Fair scheduling d) Circular wait

iv. **Inside a monitor, the number of processes that may be active at any one time is**
a) One b) Two c) Equal to the number of condition variables d) Unlimited

v. **Shared variable count is 5. A producer executes count++ and a consumer executes count– concurrently with no synchronization. The possible final values are**
a) 5 only b) 4 or 6 only c) 4, 5 or 6 d) any value between 0 and 5

vi. **Peterson's solution is designed for**
a) one process b) exactly two processes c) any number of processes d) only kernel threads

vii. **In deadlock avoidance, each process must declare in advance**
a) its priority b) the maximum number of instances of each resource type it may need
c) its expected running time d) the identity of every other process it will interact with

Group B

Attempt any four questions from the following. **[5 × 4 = 20]**

2. Write Peterson's algorithm for two processes. Show that it satisfies mutual exclusion, progress and bounded waiting, and state two limitations of the solution. **(2+3)**

3. Write the semaphore solution to the readers–writers' problem and explain briefly the role of ***readcount*** and ***mutex*** , and also discuss the starvation caused by this solution. **(5)**

4. State and explain the four necessary conditions for deadlock. Explain, with a suitable example, why all four conditions must exist simultaneously for a deadlock to occur. **(3+2)**

5. Consider the following snapshot of a system and explain whether the system is safe or not. **(5)**

Process	Allocation			Maximum Demand			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

6. Using a suitable example of a resource-allocation graph that contains a cycle but is not deadlocked. Explain precisely why the cycle does not force a deadlock in this case. **(5)**

7. Distinguish between deadlock and starvation with suitable examples and explain briefly how incorrect use of semaphore operations can lead to both deadlock and starvation. **(3+2)**

ACADEMY OF TECHNOLOGY

Internal Examination (CA2) Odd Sem, 2026

Programme/Discipline: B.TECH , CSE **Semester:** 5th

Paper Name: Introduction to Industrial Management **Paper Code:** HSMC 501 **UPID:** 5508

Date of Examination: 08/10/26 **Duration:** 1 Hour **Full Marks:**25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1.vii	2	3	4	5	6	7
CO Mapping	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2	CO2
Bloom's Level	L1	L1	L2	L1	L2	L1	L1	L1	L2	L2	L1	L3	L3

Group A

1 **Attempt any five questions from the following** [1 × 5 = 5]

i. Define event.

ii. Define activity.

iii. What is a network diagram?

iv. Define critical activity.

v. What do you understand by 'most likely time'?

vi. Define slack or float in PERT.

vii. Define dummy activity.

Group B

Attempt any four questions from the following.

2. Define optimistic, most likely time and pessimistic time, & establish the relationship between them. [5]

3. Explain the concept of crashing with one application. [5]

4. Can you provide an example of a real-world project where PERT or CPM would be beneficial in scheduling and management? [5]

5. Write the differences between PERT and CPM? [5]

6. Construct a network diagram for the project where activities(A) and predecessor(P) relationships are given below: [5]

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	-	-	A	B	B	C	D	E	H,I	F,G

7. A project consists of seven activities and the time estimates of the activities are furnished as under:

Activity	Optimistic Days	Most likely Days	Pessimistic Days
1-2	4	10	16
1-3	3	6	9
1-4	4	7	16
2-5	5	5	5
3-5	8	11	32
4-6	4	10	16
5-6	2	5	8

i. Draw the network diagram.
ii. Identify the critical path and its duration. [2+2+1]

ACADEMY OF TECHNOLOGY
Internal Examination (CA2), Odd Sem, 2026

Programme/Discipline: Computer Science and Engineering | **Semester:** 5th

Paper Name: Artificial Intelligence | **Paper Code:** PEC-IT501B | **UPID:** 005510

Date of Examination: 08/10/2026 | **Duration:** 1 Hour | **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1. i	1. ii	1. iii	1. iv	1. v	1. vi	1. vii	2	3	4	5	6	7
CO Mapping	1	1	1	1	1	1	1	1	1	1	1	1	1
Bloom's Level	1	1	1	1	1	1	1	3	2	2	2	2	2

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i. Heuristic values are admissible if

a) $h(n) > h^*(n)$ b) $h(n) <= h^*(n)$ c) $h(n) = \text{constant}$ d) none of these

ii. The time complexity of Breadth First Search is (with branching factor b and depth d)

a) $O(b^d)$ b) $O(d^b)$ c) $O(b^{d/2})$ d) $O(d^{b/2})$

iii. Which data structure is conveniently used to implement A*?

a) Stacks b) Queues c) Priority Queues d) All of the mentioned

iv. Which is not a heuristic search?

a) A* search b) Hill-climbing c) Simulated annealing d) DFS

v. Which is used to provide feedback to the learning element?

a) Critic b) Actuators c) Sensor d) None of the mentioned

vi. Heuristic function h(n) is defined as the

a) Lowest path cost b) Cheapest path from root to goal node

c) Estimated cost of cheapest path from root to goal node d) Average path cost

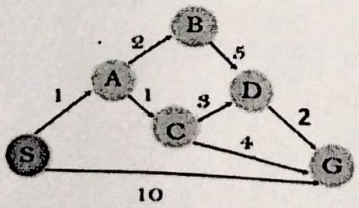
vii. The space complexity of Depth First Search is (with branching factor b, depth d and the maximum depth m)

a) O(d) b) O(bm) c) O(b) d) $O(b^m)$

Group B

Attempt any four questions from the following.

2. Apply A* search algorithm to evaluate the optimal path to reach the goal. 'S' is the start node and 'G' the goal node. [5]



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

3. Compare and contrast BFS and DFS. [5]

4. Write iterative deepening search algorithm.[5]

5. Describe Utility-based Agent with a diagram.[5]

6. You are given two jars- a 4-litre one and a 3-litre one. Neither has any measuring mark on it. There is a pump that can be used to fill the jars with water. Evaluate how you can get 2 litres of water into the 4-litre jug? With the help of a state-space diagram, describe the production rules and find a solution. [5]

7. Under which conditions A* algorithm provides an optimal solution? [5]

ACADEMY OF TECHNOLOGY
INTERNAL EXAMINATION (CA2), ODD SEMESTER, 2026

Program/Discipline: CSE | Semester: 5th
Paper Name: Software Engineering | Paper Code: ESC501 | UPID: 5505
Date of Examination: 07/10/2026 | Duration: 1 Hour | Full Marks: 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO1	CO1	CO1	CO1	CO2	CO1	CO3	CO2	CO1	CO3	CO3	CO3	CO3
Bloom's Level	L1	L1	L1	L1	L1	L1	L2	L2	L1	L3	L3	L3	L4

Group A

Attempt any five questions from the following: [1 × 5 = 5]

i) Define SRS?
ii) Slack time for critical activity is _____.
iii) Context free diagram of a system refers to which level of a DFD?
iv) Which requirements come under non-functional requirement?
v) Define data coupling?
vi) Mention the names of different sub-models of the COCOMO model.
vii) What do you mean by LOC?

Group B

Attempt any four questions from the following:

2. Justify the statement 'high cohesion and low coupling is required for good design' [5]
3. What kind of problems may be occurred without a proper SRS document? [5]
4. Draw a DFD of Root Mean Square (RMS) system with input a, b, c. [5]
5. Explain Decision Tree with proper example. [5]
6. Apply COCOMO model to find effort and development time of a product having size of 45000 LOC. [5]
7. Draw a CPM diagram and find the critical path. Also calculate ES, EF, LS and LF of activities. [5]

Task	A	B	C	D	E	F
Duration (Days)	5	4	5	6	3	4
Predecessor	-	A	A	B	C	D, E

AOT

ACADEMY OF TECHNOLOGY
INTERNAL EXAMINATION (CA2), ODD SEM, 2026

Programme/Discipline: CSE
Paper Name: Compiler Design
Date of Examination: 6/10/2026

Paper Code: PCC-CS501
Duration: 1 Hour

Semester: 5th
UPID: 5506
Full Marks: 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2	3	4	5	6	7
CO Mapping	CO3	CO3	CO1	CO3	CO2	CO1	CO3	CO3	CO3	CO3	CO3	CO3	CO3
Bloom's Level	L1	L1	L1	L1	L2	L2	L1	L3	L3	L3	L2	L3	L1

Group A: Attempt any five questions from the following [1x5=5]

1. i. "Panic mode" is a _____.
   a) Type of parser
   b) Type of top-down parser
   c) Type of bottom-up parser
   d) Error recovery technique

   ii. Which of the following derivations does a top-down parser use while parsing an input string?
   a) Leftmost derivation
   b) Leftmost derivation in reverse
   c) Rightmost derivation
   d) Rightmost derivation in reverse

   iii. Which of the following phase of the compiler is Syntax Analysis?
   a) Second
   b) Third
   c) First
   d) All the mentioned

   iv. SLR(1) is a type of_____.
   a) Top-Down Parser b) Bottom-Up Parser c) Mixed Parser d) None of the mentioned.

   v. A grammar that produces more than one parse tree for same sentence is called as_____ ?.
   a) Ambiguous
   b) Unambiguous
   c) Regular
   d) All of these

   vi. A programmer, writes a program to multiply two numbers instead of dividing them by mistake. who can detect this error?
   a) Compiler or Interpreter
   b) Compiler only
   c) Interpreter only
   d) None of the mentioned.

   vii. Parsing is categorized into how many types?
   a) three types b) four types c) two types d) five types

Group B: Attempt any four questions from the following. [4x5=20]

2. Find First() and Follow() for the following: [5]
   $S \rightarrow AB$
   $A \rightarrow a/b/\epsilon$
   $B \rightarrow c$
3. Check whether the following grammar is ambiguous or not. [5]
   $E \rightarrow E+E \mid E*E \mid id$
4. Check whether the grammar is LL(1) or not [5]
   $S \rightarrow aSbS/bSaS/\epsilon$
5. Describe Panic mode and Phrase level recovery technique with examples. [5]
6. Derive the SLR (1) parse table for the following list of productions: [3+2]
   $E \rightarrow T+E \mid T$
   $T \rightarrow id$
7. What are the types of parser in compiler design? Explain in brief. [2+3]

ACADEMY OF TECHNOLOGY

Internal Examination (CA2), Odd Sem, 2026

Discipline: CSE **Sem:** 5 **Paper Name:** Object Oriented Programming **Paper Code:** PCCCS503

UPID: 5504 **Date of Examination:** 07/10/2026 **Duration:** 1 Hour **Full Marks:** 25

Question-wise Mapping with Course Outcomes (COs) and Cognitive Levels Assessed:

Questions No	1.i	1.ii	1.iii	1.iv	1.v	1.vi	1.vii	2.	3	4.	5	6.	7.
CO Mapping	CO1	CO4	CO1	CO1	CO1	CO1	CO1	CO2, CO1	CO3	CO3	CO2	CO3	CO4
Bloom's Level	L1	L4	L1	L2	L1	L1	L2	L2, L1	L3	L3	L2	L3	L4

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

(i) Which class is the base class of all classes in Java? a) Class b) Object c) Base d) Main

(ii) Which of the following correctly declares an abstract method intended to be public?

a) public abstract void add(); b) public abstract void add() { }

c) public abstract add(); d) public virtual void add();

(iii) Which keyword can be used to prevent a class from being inherited? a) static b) abstract c) final d) private

(iv) Which of the following correctly defines an interface in Java?

a) class MyInterface { } b) abstract interface MyInterface { }

c) interface MyInterface { } d) implements MyInterface { }

(v) What is a marker interface in Java? a) An interface containing only static methods

b) An interface containing no members and used to provide special information to the Java system

c) An interface that cannot be implemented d) An interface containing only private methods

(vi) Which package is automatically imported into every Java program?

a) java.util b) java.io c) java.lang d) java.net

(vii) Which mechanism allows a subclass to provide its own implementation of a superclass method?

a) Method overloading b) Method overriding c) Constructor overloading d) Encapsulation

Group B

Attempt any four questions from the following. [5 x 4]

2. Explain the different types of inheritance supported through classes in Java with suitable diagrams. What is the purpose of the super and super() in inheritance? 3+2

3. Develop a Java program to demonstrate upcasting using a superclass reference referring to a subclass object and also to demonstrate downcasting of a superclass object to a subclass reference. 5

4. Develop a Java program to demonstrate how multiple inheritance can be achieved in Java using interfaces. 5

5. Explain with coding example why an abstract method cannot be declared as final and also explain why that cannot be declared as private. 5

6. Write a Java program using a Shape interface and classes Circle and Rectangle to demonstrate run-time polymorphism. 5

7. Analyze the accessibility of private, default, protected, and public members when the accessing class belongs to the same package , different package and subclass , non-subclass. 5